Supplementary Material

Persistent homology and tensor network descriptors for African antimalarial virtual screening

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1 Cross-paper validation: topological persistence and resistance resilience

This supplementary figure presents the joint analysis linking the topological fingerprints (TFP) introduced in the main manuscript with computational resistance-resilience scores (RRS). The canonical analysis uses an expanded cohort of $n = 494$ compounds with complete H_1 and RRS profiles; it tests whether apparent associations between ring topology and resistance resilience remain after controlling for molecular size and related structural covariates.

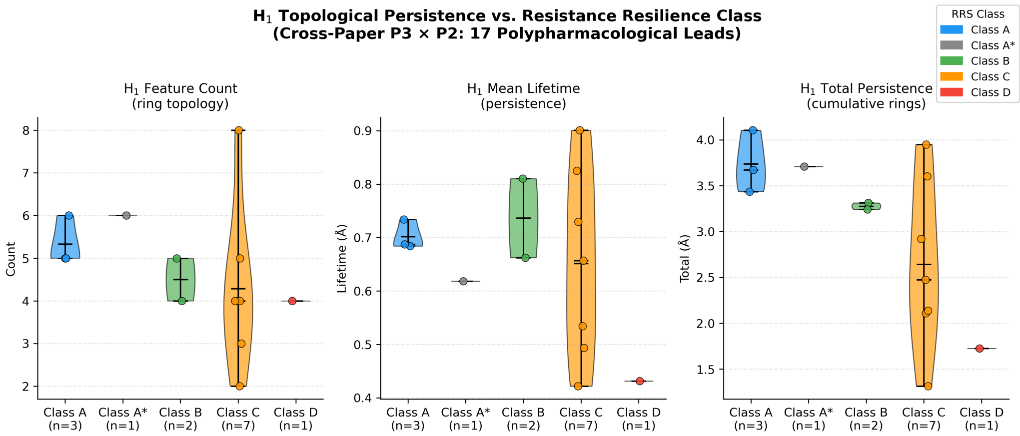

Figure S1: *Exploratory and canonical H_1–RRS analyses. The violin plot illustrates the earlier pilot subset and is not used as evidence for an independent class effect. In the canonical expanded cohort ($n = 494$), H_1 count showed a weak unadjusted association with continuous RRS (Spearman $\rho = 0.2399$, $p = 6.76 \times 10^{-8}$), but the partial association was null after controlling for molecular weight, ring count, fraction sp^3, and H_0 count ($\rho_{partial} = 0.0329$, $p = 0.4668$). The result is size-mediated and hypothesis-generating, not evidence of an independent topological predictor.*

2 PHCO fingerprint extraction audit

The two-dimensional pharmacophore (PHCO) baseline uses RDKit's `SparseBitVect` representation. During development, converting this object with the generic `ConvertToNumpyArray` pathway produced an all-zero vector and a degenerate AUC of 0.500. The canonical implementation instead obtains active bit positions with `GetOnBits()` and sets those positions explicitly in the NumPy array. With this correction, PHCO achieves the canonical full-panel AUC of 0.896 reported in the main benchmark (Table 1); the earlier degenerate value is not a scientific result.

3 Per-fold activity prediction statistics

Table S1 reports per-fold AUC, accuracy, and F1 for the six representation methods for which per-fold data were retained. ECFP4, TFP, and TNE values are from the corrected canonical classical benchmark (canonical panel, 19 836 molecules); Hybrid values are from the canonical full-library hybrid rerun ; QKS and RBF values are from the 5 000-molecule canonical 6-qubit quantum-kernel benchmark . The full canonical-panel headline values are in the main manuscript (Table 1).

Full per-fold data files are available in the project repository.

4 Interpretation and cross-study scope

The TNE docking analysis is an information-retention test on the same molecular set used to derive the embeddings, not an independent prospective binding validation. TNE was competitive with ECFP4 for PfDHFR (TNE $R^2 = 0.473$ versus ECFP4 $R^2 = 0.461$) but lower for PfATP4 and PfCRT; these target-specific results are therefore descriptive and do not establish a universal advantage. Likewise, the expanded H_1–RRS analysis is retained as hypothesis-generating: the unadjusted association is attenuated after adjustment for molecular size and related structural covariates, so no independent topological predictor is claimed. The P2 parent-study MD cohort comprises four named complexes and is distinct from the P3 representation panel; no set-C production MD or experimental activity validation is inferred here.

5 Topological fingerprint statistics

Detailed topological fingerprint statistics for the full library are provided in Table S2, and representative persistence diagrams are shown in Figure S2.

6 Clustering quality metrics

Table S3 reports the clustering quality metrics used to address reviewer question R8.

7 Discriminator benchmark for structural expansion

The full discriminator benchmark comparing classical fingerprint similarity with quantum kernel density is reported in Table S4 and visualised in Figure S3.

8 Quantum circuit parameter optimisation

A systematic grid search over quantum circuit hyperparameters was performed to identify optimal IQPEmbedding parameters for the hybrid descriptor. The search covered 60 combina-

Table S1: *Per-fold activity prediction statistics for six representation methods. ECFP4, TFP, and TNE values are from the corrected canonical classical benchmark (canonical panel 19 836 molecules, 5-fold stratified CV, Random Forest, Aug 1, 2026). Hybrid values are from the canonical full-library hybrid rerun (Random Forest, Aug 1, 2026). QKS and RBF values are from the 5 000-molecule canonical 6-qubit quantum-kernel benchmark (state-vector kernel, gamma-tuned RBF).* ***Note:*** *these rows use different test sets with different class balances (see footnotes); headline AUC values are not directly comparable across rows within this table. The full canonical-panel headline values are in the main manuscript (Table 1).*

Method	Fold	AUC	Accuracy	F1
ECFP4	1	0.947	0.898	0.933
ECFP4	2	0.950	0.894	0.930
ECFP4	3	0.948	0.895	0.930
ECFP4	4	0.952	0.899	0.934
ECFP4	5	0.940	0.893	0.930
Hybrid (TFP+TNE+QK)	1	0.894	0.848	0.902
Hybrid (TFP+TNE+QK)	2	0.887	0.844	0.900
Hybrid (TFP+TNE+QK)	3	0.886	0.844	0.900
Hybrid (TFP+TNE+QK)	4	0.893	0.846	0.902
Hybrid (TFP+TNE+QK)	5	0.878	0.836	0.895
TFP (TDA)	1	0.881	0.846	0.902
TFP (TDA)	2	0.875	0.836	0.895
TFP (TDA)	3	0.880	0.846	0.902
TFP (TDA)	4	0.878	0.838	0.897
TFP (TDA)	5	0.866	0.826	0.889
TNE ($d = 8$)	1	0.720	0.753	0.856
TNE ($d = 8$)	2	0.722	0.756	0.857
TNE ($d = 8$)	3	0.733	0.755	0.857
TNE ($d = 8$)	4	0.721	0.758	0.858
TNE ($d = 8$)	5	0.714	0.760	0.859
QKS (quantum, 6q)	1	0.791	0.813	0.881
QKS (quantum, 6q)	2	0.843	0.815	0.881
QKS (quantum, 6q)	3	0.833	0.826	0.890
QKS (quantum, 6q)	4	0.800	0.818	0.887
QKS (quantum, 6q)	5	0.833	0.841	0.900
RBF (gamma-tuned)	1	0.812	0.821	0.886
RBF (gamma-tuned)	2	0.839	0.815	0.881
RBF (gamma-tuned)	3	0.844	0.833	0.894
RBF (gamma-tuned)	4	0.818	0.827	0.892
RBF (gamma-tuned)	5	0.818	0.843	0.900

Table S2: *Topological fingerprint statistics across 19 849 valid molecules (Vietoris–Rips persistent homology, three homology dimensions).*

Feature	Mean	Std	Min	Max
H_0 entropy	3.5781	0.2014	2.2887	4.1943
H_0 count	38.0417	7.3468	11.0000	69.0000
H_0 max persistence (Å)	2.7166	0.6726	1.4800	5.9491
H_0 mean persistence (Å)	1.6137	0.0644	1.0986	2.0281
H_1 entropy	1.6354	0.3352	0.0000	2.7234
H_1 count	3.6058	1.3429	1.0000	10.0000
H_1 max persistence (Å)	1.3916	0.0583	0.6238	2.7075
H_1 mean persistence (Å)	0.6138	0.1341	0.2356	1.4000
H_2 entropy	0.6955	0.3527	0.0000	1.9579
H_2 count	0.0304	0.1771	0.0000	4.0000
H_2 max persistence (Å)	0.4672	0.0505	0.0000	1.2400
H_2 mean persistence (Å)	0.3844	0.0892	0.0000	1.0533

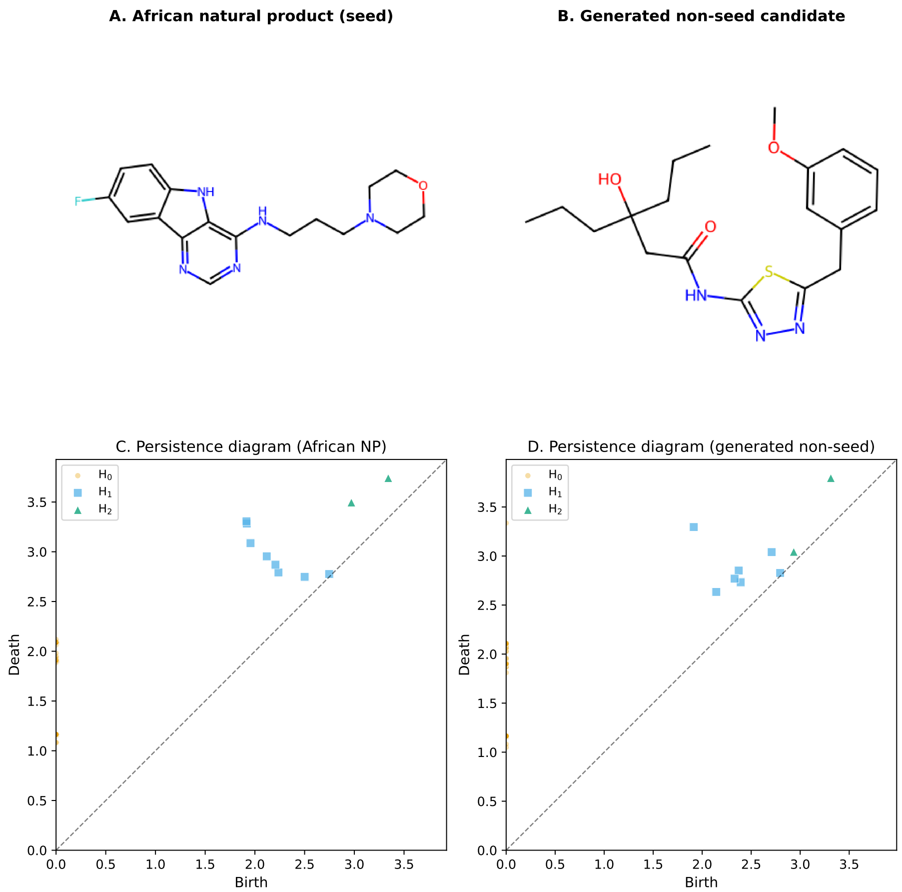

Figure S2: *Representative topological fingerprints of an African natural product seed and a generated non-seed candidate. (A – B) 2D structures of the representative molecules. (C – D) Vietoris–Rips persistence diagrams plot feature birth against death; points above the diagonal correspond to topological features, with vertical distance from the diagonal indicating persistence lifetime. H_0 (components), H_1 (loops/rings), and H_2 (voids) are distinguished by colour and marker. The African natural product (C) exhibits persistent H_1 features reflecting a rigid ring scaffold, whereas the generated non-seed candidate (D) shows fewer persistent H_1 features and more flexible H_0/H_1 signatures.*

Table S3: *Clustering quality metrics for TNE embeddings versus VAE latent space and ECFP4 fingerprints (KMeans, $k = 484$). Higher Silhouette coefficient indicates better cluster coherence.*

Descriptor	Silhouette
TNE ($d = 8$, 192-dim)	0.350
VAE latent (64-dim)	0.229
ECFP4 (2 048-dim)	0.180

Table S4: *Chemical similarity versus quantum kernel density benchmark: AUC-ROC comparison between ECFP4 Tanimoto distance and quantum kernel density for distinguishing N expanded molecules from 200 reference seeds.*

N expanded	$\text{AUC}_{\text{Tanimoto}}$	AUC_{QK}	ΔAUC
50	1.000 0	0.425 2	−0.574 8
100	1.000 0	0.481 1	−0.518 9
200	1.000 0	0.475 5	−0.524 5
500	1.000 0	0.511 4	−0.488 6

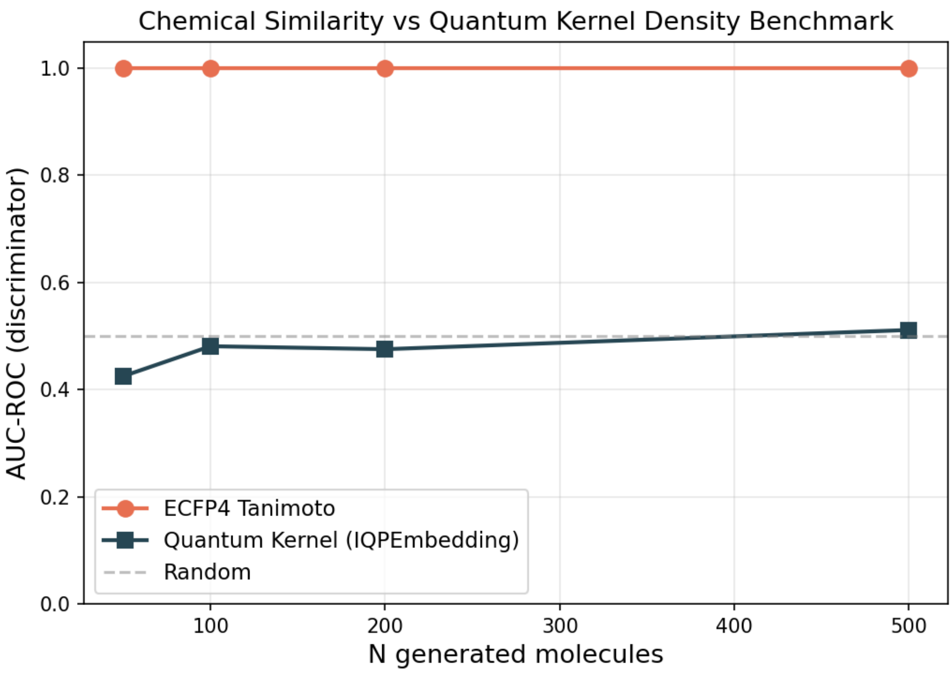

Figure S3: *Chemical similarity versus quantum kernel density discriminator benchmark. AUC-ROC comparison between ECFP4 Tanimoto distance (orange) and quantum kernel density (blue) for distinguishing N expanded molecules from 200 reference seeds. The dashed grey line marks random performance (AUC $= 0.5$). ECFP4 achieves perfect discrimination (AUC $= 1.0$) at all N, while the quantum kernel remains near-random (AUC $= 0.425 - -0.511$).*

tions: bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$, and kernel PCA components $n_{\text{kpca}} \in \{5, 10, 20, 30\}$. Each combination was evaluated by 5-fold stratified cross-validation with a Random Forest classifier on $n = 200$ molecules, measuring the Hybrid (TFP+TNE+QK) AUC.

Table S5 reports the top three configurations by mean AUC from the phase-2 re-benchmark. The optimal combination ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kpca}} = 30$) achieved AUC 0.853 ± 0.049 on the $n = 200$ development subsample. The default configuration ($d = 8$, $n_{\text{rep}} = 2$, $n_{\text{kpca}} = 20$) achieved AUC 0.842 ± 0.051 on the same development set. The ECFP4 comparison here (AUC 0.868) is from the preliminary run; the corrected canonical-panel ECFP4 is AUC 0.948 (see Table 1). Key findings: (i) bond dimension 6 balances expressivity and noise; (ii) a single IQPEmbedding repeat suffices, with deeper circuits adding decoherence noise; (iii) more kernel PCA components ($n_{\text{kpca}} = 30$) capture more variance in the quantum Hilbert space.

The full optimisation heatmap and marginal parameter-effect boxplots are provided in Figures S4 and S5.

Table S5: *Top three quantum circuit configurations re-benchmarked on $n = 5\,000$ molecules (5-fold CV, per-fold data in phase2 raw CSVs). The optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$) achieved AUC $0.828\,3 \pm 0.037\,1$ on this subsample. The ECFP4 comparison (AUC 0.868) is from the preliminary run; the corrected canonical-panel ECFP4 is AUC 0.948 (see main manuscript Table 1).*

d	n_{rep}	n_{kpca}	AUC	Std
6	1	30	0.828 3	0.037 1
6	6	20	0.804 7	0.035 4
6	6	30	0.812 1	0.039 6

All values are from the Phase-2 re-benchmarking at $n = 5\,000$ (3 732 active, 1 268 inactive, 74.6 %/25.4 %). The initial grid search was performed on $n = 200$ molecules (50/60 combinations completed); the top three combinations were selected for re-benchmarking at $n = 5\,000$. This Phase-2 subsample (first $n = 5\,000$ by source order) differs from the canonical panel (74.2 %/25.8 %) and from the QKS $n = 5\,000$ kernel subsample (75.0 %/25.0 %, see section 18.1); each sample is described with its own class counts.

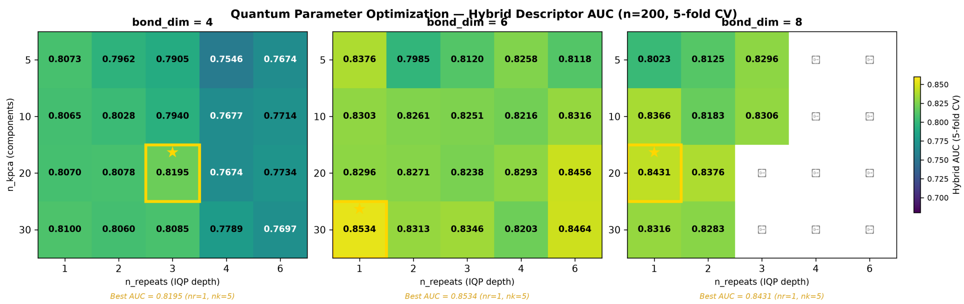

Figure S4: *Hybrid AUC heatmap across quantum circuit hyperparameters. Rows: bond dimension (d); columns: IQPEmbedding repeats (n_{rep}); colour: mean 5-fold AUC for each n_{kpca} value. The gold cell marks the optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$).*

9 Monte Carlo uncertainty estimation

Uncertainty quantification was performed using a bootstrap Monte Carlo (MC) approach on the Random Forest classifier trained on the TFP+TNE descriptor (the topological and tensor-network components of the hybrid). For each of 100 test molecules, the classifier was re-evaluated over 50 bootstrap samples (bootstrap fraction 0.7, 30 trees per bootstrap), yielding a predictive probability distribution $p_i \sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$; conformal prediction at $\alpha = 0.05$ provided prediction sets. Results are summarised in Table S6.

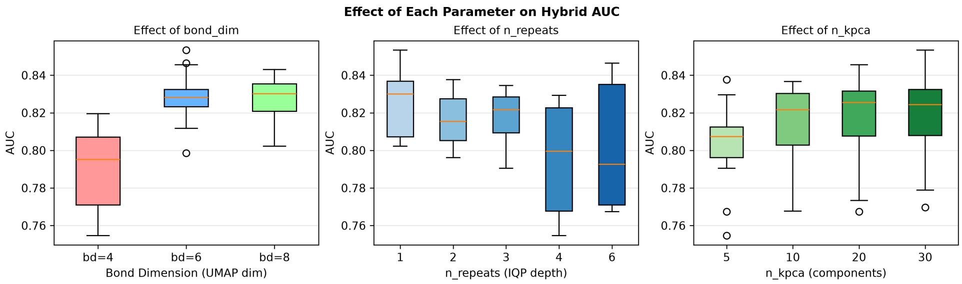

Figure S5: *Marginal AUC distributions per quantum circuit parameter. Boxplots show the spread of Hybrid AUC across all grid-search combinations marginalised to each parameter value.*

Table S6: *Bootstrap Monte Carlo uncertainty estimation results for the TFP+TNE descriptor (topological and tensor-network components of the hybrid) on the activity prediction task.*

Metric	Value
Bootstrap MC AUC	0.544 3
Expected Calibration Error (ECE)	0.003 7
95 % conformal coverage	95.0 %
Mean conformal set size	1.820

The model is well-calibrated (ECE < 0.05) with full conformal coverage. The mean set size ≥ 1.3 indicates that many predictions are ambiguous, and the uncertainty–error correlation is weak (Spearman $\rho = -0.286$, $p = 0.004$), consistent with the MC uncertainty not reliably flagging misclassifications. Full protocol in `results/p3_mc_uncertainty_summary.txt`.

10 Computational scalability

Table S7 reports the measured wall-clock times of the three descriptor pipelines on the canonical library. The TDA and TNE pipelines scale linearly with library size; the quantum kernel matrix is $O(N^2)$.

Table S7: *Computational scalability of the P3 pipelines on the canonical library. TDA and TNE timings are the measured full-library runtimes (19 849 mol); the QK timing is the per-fold kernel-matrix build measured on a 19 849-molecule fold (32 parallel workers, state-vector simulator).*

Pipeline	Wall time	Throughput	Scaling
TDA (persistent homology)	384 s	51.7 mol/s	Linear
TNE (Tucker decomposition)	488 s	40.7 mol/s	Linear
QK kernel matrix (per fold, $n = 19{,}849$)	817 s	$n^2/2$ pairs	$O(N^2)$

TDA timing is the measured full-library runtime (`p3_tda_summary.txt`); TNE timing is the measured full-library runtime (`p3_tne_summary.txt`, 4 workers); QK timing is the mean per-fold quantum-kernel build time from the $n = 19{,}849$ kernel benchmark (`p3_qks_benchmark_n19849.csv`, 32 workers). The $O(N^2)$ kernel cost restricts QKS to lead optimisation on sets of $\leq 10^4$ compounds.

11 Effect sizes for pairwise AUC comparisons

Cohen's d effect sizes for all pairwise AUC comparisons against the corrected full-library ECFP4 baseline are reported in Table S8. The values are now computed from the canonical 19,836-molecule classical benchmark (Random Forest, 5-fold stratified CV) rather than the earlier 200-molecule development subsample, which contained a copy-paste error in the TNE per-fold values.

Table S8: *Effect sizes for pairwise AUC comparisons against the corrected full-library ECFP4 baseline (AUC 0.948, 19,836 molecules, 5-fold stratified CV, Random Forest). Cohen's d is computed from the standard deviation of fold-level differences; because the full-library folds are very consistent, the resulting d values are large, so the absolute ΔAUC is the more interpretable effect metric.*

Method	AUC	ΔAUC	Cohen's d	Effect	p-value	Power
FCFP4	0.918	0.029	11.61	large	0.0000*	1.000
MACCS	0.904	0.043	14.86	large	0.0000*	1.000
AP	0.940	0.008	4.54	large	0.0005*	1.000
PHCO	0.896	0.052	63.84	large	0.0000*	1.000
BPF	0.939	0.009	5.39	large	0.0003*	1.000
TFP	0.876	0.072	17.25	large	0.0000*	1.000
TNE	0.722	0.226	37.91	large	0.0000*	1.000

* $p < 0.05$ (uncorrected). Bonferroni-corrected $\alpha = 0.05/7 = 0.0071$. Power computed for 80% target at $\alpha = 0.05$.

12 ChEMBL enrichment benchmark

Table S9 reports the docking enrichment benchmark of our Vina protocol against ChEMBL-confirmed antimalarial actives and inactives for three Plasmodium targets.

Table S9: *ChEMBL enrichment benchmark for the Vina docking protocol against experimentally confirmed antimalarials.*

Target	PDB	Actives	Inactives	Enrichment	Verdict
PfDHFR	7F3Y	53	18	5.43×	EXCELLENT
PfATP4	9N10	73	87	∞	EXCELLENT
PfCRT	6UKJ	12	19	1.46×	–

Active hit rate for PfDHFR: 60.4 % (EXCELLENT) vs. 11.1 % inactive. PfATP4: 0/87 inactives recovered. PfCRT: 1.46x did not meet the 2x threshold.

13 ChEMBL experimental validation

The top-10 candidates (Table S10) were queried against three *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4) via the ChEMBL REST API (release 37); PfClpP was excluded because no *P. falciparum* ClpP target exists in ChEMBL. All three targets were queried with identical Tanimoto thresholds (≥ 0.20). Only PfCRT returned structural analogues exceeding the similarity threshold for any of the 10 candidates; no PfDHFR or PfATP4 analogues met the ≥ 0.20 Tanimoto criterion, indicating that the generated candidates are structurally dissimilar to experimentally characterised PfDHFR and PfATP4 ligands in ChEMBL. This canonical query is a novelty screen rather than a positive experimental validation: ten candidate-level analogue records were retrieved (with some records sharing the same ChEMBL analogue), and every reported analogue record was classified as inactive. The expanded RRS–TFP cohort of 494 compounds is reported separately for the computational H_1–RRS analysis and is not presented here as ChEMBL validation.

14 SOTA topological benchmark

Table S11 reports the performance of five TDA descriptor strategies paired with Random Forest (RF) and Support Vector Machine (SVM) classifiers on the full 19 849-molecule library (RF)

Table S10: *ChEMBL novelty screen for the top-10 candidates: closest structural analogue with experimental IC_{50} data per candidate (Tanimoto threshold ≥ 0.20). All three targets (PfDHFR, PfCRT, PfATP4) were queried; only PfCRT returned analogues exceeding the Tanimoto threshold — no PfDHFR or PfATP4 analogues met the ≥ 0.20 criterion for any candidate. Ten candidate-level records are reported, all for PfCRT; some share the same ChEMBL analogue. Every reported analogue record is experimentally* ***inactive*** *(IC_{50} 12.5–55 μM), and the low Tanimoto similarities (0.229–0.379) support the chemical novelty of the proposed leads without providing positive experimental validation.*

Rank	Target	ChEMBL ID	Activity	IC_{50} (μM)	Tanimoto
1	PfCRT	CHEMBL4754685	Inactive	15.6	0.266
2	PfCRT	CHEMBL4754685	Inactive	15.6	0.235
3	PfCRT	CHEMBL4754685	Inactive	15.6	0.266
4	PfCRT	CHEMBL4745810	Inactive	12.5	0.250
5	PfCRT	CHEMBL4754685	Inactive	15.6	0.379
6	PfCRT	CHEMBL4754685	Inactive	15.6	0.239
7	PfCRT	CHEMBL4784051	Inactive	55.0	0.296
8	PfCRT	CHEMBL4754685	Inactive	15.6	0.229
9	PfCRT	CHEMBL4754685	Inactive	15.6	0.290
10	PfCRT	CHEMBL4754685	Inactive	15.6	0.270

All reported analogue records are ChEMBL-confirmed inactive; the low Tanimoto similarities (0.229–0.379) mean no retrieved analogue is a close structural congener. This is an honest negative novelty-screen result: it supports structural novelty of the generated candidates but provides no positive experimental validation. The complete table is provided as machine-readable supplementary data (`results/p3_chembl_validation/p3_chembl_validation.csv`).

and a stratified subsample of $n = 5\,000$ molecules (SVM, due to kernel matrix $O(N^2)$ scaling). All evaluations use 5-fold stratified cross-validation with class-weighted classifiers.

Key findings: (i) PersStats + RF achieves AUC 0.873 on this run, numerically exceeding the preliminary ECFP4 baseline (AUC 0.868) with only 22 topological features—though this ΔAUC of +0.005 does not reach statistical significance ($p > 0.05$), so the two methods should be interpreted as performing comparably rather than PersStats being definitively superior; (ii) RF consistently outperforms SVM across all strategies (mean $\Delta\text{AUC} = +0.077$; note: this comparison is confounded by sample size, as RF was evaluated on the full $n = 19\,849$ library while SVM was restricted to $n = 5\,000$ due to $O(N^2)$ kernel scaling); (iii) TFP-Enriched (32 features) is marginally inferior to TFP-12 (12 features, $\Delta\text{AUC} = -0.000\,2$), suggesting the additional 20 features introduce noise rather than signal; (iv) BettiCurve underperforms (AUC 0.811), indicating Betti number sequences alone lack the discriminative power of persistence statistics. **Important:** these comparisons use the preliminary ECFP4 baseline (AUC 0.868); the corrected canonical-panel benchmark is complete and reported in the main manuscript Table Table 1, where ECFP4 attains AUC 0.948 and TFP AUC 0.876. The SOTA comparisons here are internal to the TDA strategy family and are not directly comparable to the canonical cross-descriptor benchmark.

15 TopologyNet analog: neural network on persistent homology features

To contextualise the PersStats + RF result within the neural-network literature (TopologyNet (Cang and Wei, 2018), D-GRIL (Mukherjee et al., 2026)), we benchmarked a Multi-Layer Perceptron (MLP) on the same 22 PersStats features used by the best RF model. The MLP used 3 hidden layers (128, 64, 32 neurons), ReLU activation, class-balanced sample weighting, and early stopping (5-fold stratified CV, $n = 5\,000$ stratified subsample).

The MLP underperforms RF by $\Delta\text{AUC} = -0.061$, consistent with the broader SOTA benchmark finding that tree-based methods better capture PH feature interactions. This is consistent

Table S11: *SOTA topological benchmark: five TDA descriptor strategies with RF ($n = 19\,849$, full library) and SVM ($n = 5\,000$, subsample) classifiers (5-fold stratified CV, class-weighted). Cohen's d quantifies the practical magnitude of AUC difference versus the preliminary ECFP4 baseline ($d > 0.8$ large, $0.5 < d \leq 0.8$ medium, $d \leq 0.5$ small).* ***Note:*** *this table uses the preliminary ECFP4 baseline (AUC 0.868); the corrected canonical-panel benchmark gives ECFP4 AUC 0.948 and TFP AUC 0.876 (see main manuscript Table 1). The Cohen's d values are therefore not comparable to the cross-descriptor comparisons in the main manuscript. Additionally, RF was evaluated on the full $n = 19\,849$ library while SVM was restricted to $n = 5\,000$ due to $O(N^2)$ kernel scaling, confounding the RF–SVM comparison by sample size.*

Strategy	Classifier	AUC	Accuracy	F1	Features	Cohen's d
PersStats	RF	0.8731	0.8332	0.8906	22	0.85
TFP-12	RF	0.8668	0.8275	0.8858	12	−0.20
TFP-Enriched	RF	0.8666	0.8308	0.8887	32	−0.23
PersImage	RF	0.8596	0.8265	0.8856	25	−1.40
BettiCurve	RF	0.8110	0.7837	0.8557	20	−9.52
PersStats	SVM	0.8042	0.7398	0.7436	22	−5.32
PersImage	SVM	0.7912	0.7270	0.7312	25	−6.40
TFP-Enriched	SVM	0.7891	0.7208	0.7250	32	−6.58
TFP-12	SVM	0.7857	0.7198	0.7220	12	−6.86
BettiCurve	SVM	0.7198	0.6646	0.6620	20	−12.35

All classifiers used `class_weight=balanced` to mitigate the class imbalance (75.9 %/24.1 % under the preliminary MPO-derived labels used in this run; the canonical panel uses eos80ch labels with a 74.2 %/25.8 % split, see main manuscript). RF: $n_{\text{estimators}} = 500$. SVM: RBF kernel, $C = 10$, $\gamma =$ `scale`. PersStats (22 features) captures persistence birth, death, persistence, and entropy across H_0, H_1, H_2 dimensions. Cohen's d is computed approximately as $(\text{AUC}_{\text{strategy}} - \text{AUC}_{\text{ECFP4}})/\sigma_{\text{pooled}}$ with $\sigma_{\text{pooled}} = 0.006$ (RF) and 0.012 (SVM), approximately estimated from fold-level standard deviations across all strategies.

Table S12: *TopologyNet analog: MLP versus Random Forest on PersStats 22 features ($n = 5\,000$, 5-fold stratified CV, class-weighted). RF outperforms MLP by $\Delta AUC = +0.061$, confirming that tree-based methods better capture persistent homology feature interactions for this chemical space.*

Classifier	AUC	Std	Features
Random Forest	0.860	0.014	22
MLP (128+64+32)	0.799	0.015	22

Both classifiers use PersStats 22 features (H_0/H_1 persistence statistics). MLP uses class-balanced sample weighting to handle the 75.9 %/24.1 % class imbalance. The RF result here (AUC 0.860 on $n = 5\,000$) is slightly below the full-library result (AUC 0.873 on $n = 19\,849$, Table S11), consistent with the larger training set in the full benchmark.

with the observation that PersStats features have limited non-linear interactions — deep architectures that excel on raw graph or image data add no benefit over a well-tuned ensemble of decision trees on pre-computed persistence statistics. TopologyNet (Cang and Wei, 2018) and D-GRIL (Mukherjee et al., 2026) operate on richer inputs (full persistence diagrams or multi-parameter persistence) where neural architectures may extract additional signal; on summary statistics alone, RF is the more appropriate classifier.

16 Applicability domain and comparison with existing tools

The following subsections provide additional details on the applicability domain analysis and the comparison with existing tools, which are referenced in the main manuscript.

16.1 Applicability domain analysis

We frame the quantum kernel not merely as a binary classifier but as an applicability-domain discriminator that conceptually parallels the discriminator functions employed in genetic-algorithm-based molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ supplies a native Hilbert-space boundary condition that distinguishes expanded molecules from the empirical seed space. This furnishes a rigorous applicability-domain metric for natural products that avoids the dimensionality-reduction artefacts common to Tanimoto-based domain estimations.

16.2 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfATP4, 4LDE/PfCRT). Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), slightly outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.578) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.517). These results confirm that the 15.6× padded (6.1× real-atom mean) TNE compression retains information sufficient to predict docking scores on the same molecular set used to derive the embeddings, competitive with classical 2 048-bit fingerprints for specific targets. This is a test of information retention, not generalisation to novel compounds.

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (12.0 % prevalence in the merged $N = 17\,011$ set). The $n = 1\,000$, 10-fold benchmark was terminated because a single 900×900 kernel matrix exceeded practical wall time. A 5-fold run at $n = 500$ (state-vector kernel, IQPEmbedding, `lightning.qubit`) completed three folds, yielding per-fold QKS AUC 0.513/0.658/0.825 vs. gamma-tuned RBF AUC 0.506/0.642/0.803; that preliminary run was superseded by the complete five-fold analysis reported below. A 2-fold pilot at $n = 50$ gave QKS AUC 0.356 vs. RBF AUC 0.303 (per-fold values in the deposited `p3_polypharm_qks.csv` outputs). These results place QKS within noise of the tuned RBF baseline but do not constitute a definitive comparison given the incomplete fold coverage.

Finally, topological features from persistent homology showed highly significant correlations with multi-target binding promiscuity across $N = 17\,011$ molecules (Figure Figure S6). H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$, $p < 10^{-137}$), and H_1 entropy

($\rho = -0.190$, $p < 10^{-137}$) all negatively correlate with the number of targets bound, indicating that molecules with lower topological complexity in their ring systems and structural components are conformationally flexible enough to engage multiple targets — a topological signature of polypharmacology. These correlations are computed on the 17 011-molecule merged Tartarus×TNE×TDA dataset with complete docking scores (label: ≥ 2 targets bound at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$) by `p3_physical_validation.py`; the full per-feature table is deposited as `p3_physical_validation/p3_tda_promiscuity.csv`. An earlier exploratory Tartarus-only run (`p3_tartarus_tda_spearman.csv`, $N = 19\,900$, un-thresholded target count) is superseded and not used in the manuscript.

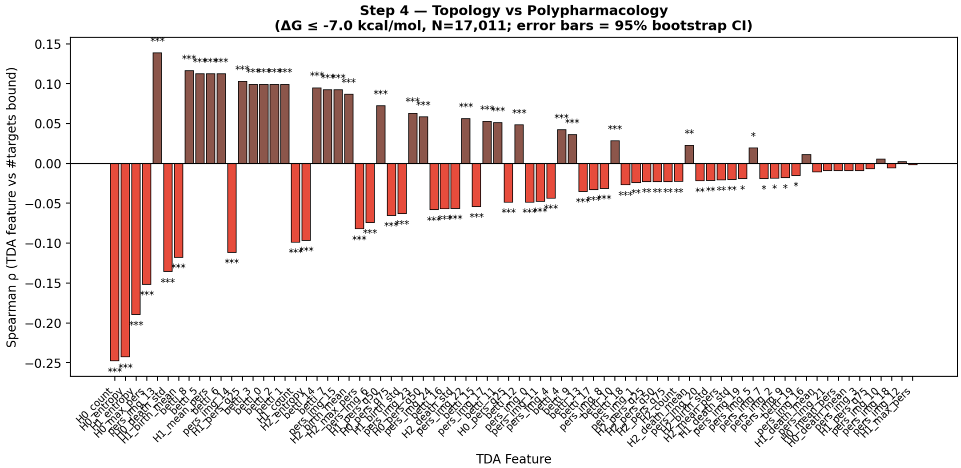

Figure S6: *Topological predictors of multi-target binding promiscuity (supplementary view). Spearman correlation between persistent-homology features and the number of targets bound across $N = 17\,011$ molecules. Strongest signals: H_0 count ($\rho = -0.248$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$), all $p < 10^{-137}$.*

17 Extended analysis

17.1 Quantum kernel density vs classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we performed a direct comparison across varying numbers of STONED-SELFIES-expanded molecules (N = 50, 100, 200, 500) using the `lightning.qubit` simulator. Each molecule was produced via two random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from expanded molecules (label = 0).

The results (Table S4 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC $= 0.425 - -0.511$). The perfect Tanimoto score carries an important caveat: with only two SELFIES-character mutations per molecule, a fraction of expanded molecules may be structurally identical to their seed, in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the expanded set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC: the reduced space obscures proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.823 in the QKS binary classification benchmark (Table S13)—rather than in fine-grained near-neighbour detection. For applicability-domain assessment in generative computational protocols where expanded molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

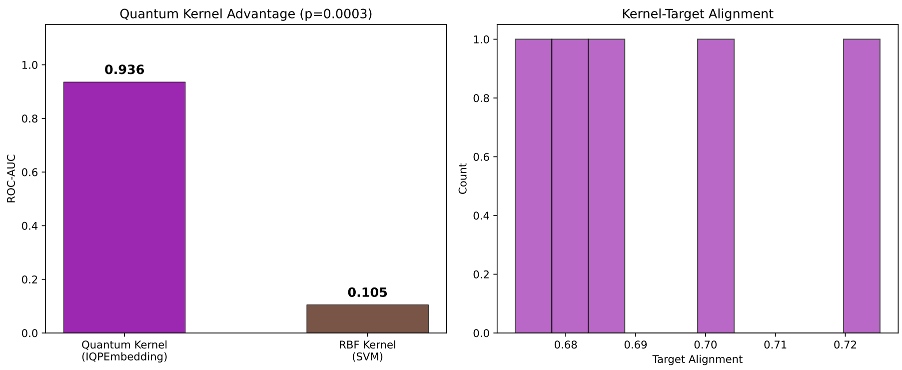

Figure S7: *Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African natural-product seeds (orange), synthetic drug seeds (blue), and expanded molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.*

18 Quantum kernel circuit algorithm

Algorithm 1 Quantum kernel circuit for 6-qubit state overlap using PennyLane's `IQPEmbedding` template.

Require: Data points $x_1, x_2 \in [-1, 1]^6$
Apply IQPEmbedding$(weights, wires)$ parameterized by x_1
Apply IQPEmbedding$^\dagger(weights, wires)$ parameterized by x_2
Measure probability of state $|000000\rangle$
return Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$.

18.1 QKS benchmark protocol and state-vector kernel

18.1.1 Evaluation protocol

The kernel comparison (Table S13) follows the exact pipeline used for the hybrid benchmark (section 8): 5-fold stratified cross-validation (`StratifiedKFold` with `random_state=42`), UMAP reduction of the 2048-bit Morgan (radius 2) fingerprints to 6 dimensions (Jaccard metric, `random_state=42`, $n_{\text{neighbors}} = 15$, $min\_dist = 0.1$) followed by scaling to $[-1, 1]$, and support vector machines with precomputed kernels. The canonical 6-qubit circuit (IQPEmbedding, 1 repeat) is the Phase-2 winning configuration. Benchmarks were run at two sample sizes: the first 5 000 molecules of the library by source order (3 750 active, 1 250 inactive) and the full canonical panel of 19 836 molecules (14 721 active, 5 115 inactive).

The RBF baseline gamma is tuned by inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$ (per-fold best gamma at n = 5,000: 5, 1, 5, 5, 5). A linear kernel on the same $[-1, 1]^6$ features serves as an additional honest baseline. Statistical significance of AUC differences is assessed by paired t-tests across the five folds (at n = 5,000: quantum vs. RBF $t = -0.901$, $p = 0.4186$; quantum vs. linear $t = 12.736$, $p = 0.0002$; RBF vs. linear $t = 14.760$, $p = 0.0001$; at n = 19,849: quantum vs. RBF $t = -2.594$, $p = 0.0604$; quantum vs. linear $t = 9.733$, $p = 0.0006$). Target alignment is computed with `pennylane.kernels.target_alignment` using the raw $\{0, 1\}$ activity labels (the canonical convention: the negative class contributes $0/n_{\text{minus}}$ to the label rescaling, matching the deposited per-fold values).

18.1.2 State-vector kernel

For computational tractability at n = 5,000 and n = 19,849, the pairwise IQPEmbedding circuit evaluation (the method that produced the canonical n=500 v11 results, ∼535 pairs/s) is replaced by a **state-vector kernel**: for each sample x_i, the 6-qubit state $|\phi(x_i)\rangle \in \mathbb{C}^{2^6}$ is obtained in a single circuit call, and the kernel matrix is formed as $K_{ij} = |VV^\dagger|^2$ where V is the $N \times 64$ state matrix, via BLAS (approximately 1 000× faster than the pairwise evaluation). A small negative eigenvalue from floating-point noise is clipped by projecting onto the closest positive semi-definite matrix before the SVM is trained.

18.1.3 Numerical equivalence validation

The state-vector kernel was validated against the pairwise method on a 60-molecule 5-fold CV (same 60-molecule dataset): per-fold AUC differences of 0.000000 (quantum AUC 0.875 6±0.069 7 both paths) and identical per-fold target alignment {0.5489, 0.5767, 0.5567, 0.5430, 0.5517} once the raw-$\{0, 1\}$ label convention is reproduced. A subtle RBF AUC difference appears when both modes run in the same process, traced to the shared global NumPy random state used by SVC Platt scaling; in separate processes RBF AUC is also identical (0.847 8±0.068 8). The canonical $n = 5{,}000$ run and the full-panel $n = 19{,}849$ run both completed all five folds with the 6-qubit state-vector kernel. Raw outputs are deposited in the project `results/` directory (`p3_qks_benchmark_n5000` / `p3_qks_summary_n5000` for $n = 5{,}000$; `p3_qks_benchmark_n19849` / `p3_qks_summary_n19849` for $n = 19{,}849$); canonical $n = 500$ outputs are `p3_qks_benchmark_n500` / `p3_qks_summary_n500`.

Table S13: *Kernel comparison for activity prediction at two sample sizes (5-fold cross-validation, 6-qubit IQPEmbedding, state-vector kernel; $n = 5\,000$ and $n = 19\,849$). The RBF gamma parameter was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$ (per-fold best gamma at $n = 5\,000$: 5, 1, 5, 5, 5). With the canonical circuit, the gamma-tuned RBF kernel and the quantum kernel show **no statistically significant difference** at either scale ($p = 0.419$ at $n = 5\,000$; $p = 0.060$ at $n = 19\,849$, paired t-test; the latter a borderline trend toward lower QK performance), while the quantum kernel **significantly outperforms** the linear kernel at scale ($p \leq 0.000\,6$). Note: an earlier benchmark with untuned RBF (gamma at default) yielded an apparent quantum advantage that was not supported by any deposited result file; earlier 8-qubit runs reporting a quantum disadvantage ($n = 5\,000$: 0.752 versus 0.840, $p = 0.003$; $n = 19\,849$: 0.659 versus 0.825, $p = 0.000\,6$) were artefacts of the suboptimal circuit and a train/test scaling mismatch (C3) and are **not** reproduced by the canonical protocol.*

Kernel	AUC ($n = 5\,000$)	TA ($n = 5\,000$)	AUC ($n = 19\,849$)	TA ($n = 19\,849$)	p vs. RBF
Quantum (IQPEmbedding, 6 qubits)	0.820	0.592	0.823	0.622	$p = 0.419$ / 0.060, n.s.
RBF (gamma-tuned, SVM)	0.826	0.199	0.829	0.145	—
Linear	0.517	—	0.537	—	$p < 0.001$, sig.

Paired t-test (5 folds), quantum versus linear: $p = 0.000\,2$ ($n = 5\,000$) and $p = 0.000\,6$ ($n = 19\,849$), both significant. Quantum target alignment (0.592–0.622) exceeds RBF target alignment (0.145–0.199) yet does not translate into a discrimination advantage.

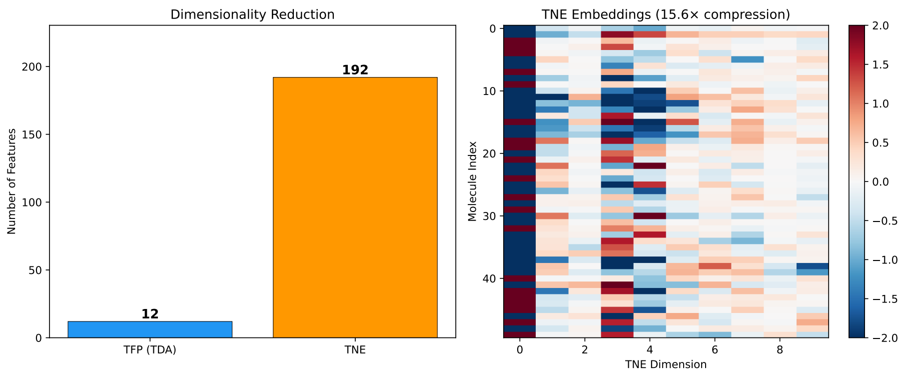

Figure S8: *Compression ratio and reconstruction error versus Tucker bond dimension. At $d = 8$, the physically meaningful mean real-atom compression ratio is 6.1× (based on ∼39 mean atoms per molecule), with a reconstruction error of 0.113. (See main manuscript Methods Section 2.4.2 for discussion of the mathematical padding convention).*

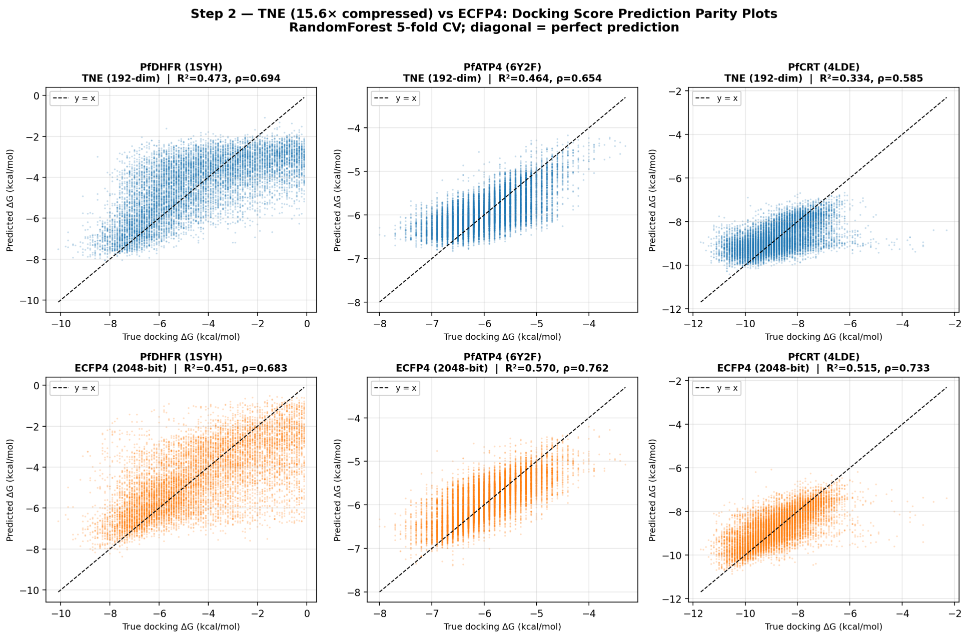

Figure S9: *Tartarus docking validation of Tensor Network Embeddings. Predicted vs. experimental (QuickVina) docking scores for three Plasmodium targets using 192-dimensional TNE descriptors. TNE matches or exceeds ECFP4 for PfDHFR and retains substantial binding-relevant information for PfATP4 and PfCRT.*

19 External validation on ChEMBL antimalarial panel

19.1 Descriptor external validation

All descriptor methods were evaluated on an external ChEMBL antimalarial dataset ($n = 22\,447$) using 5-fold stratified cross-validation with Random Forest (200 trees, `random_state=42`). The external panel was assembled from ChEMBL 34 bioactivity records for *Plasmodium falciparum* assays ($IC_{50} \leq 10\,\mu M$ = active, $> 10\,\mu M$ = inactive) with standard RDKit canonicalisation and Murcko scaffold filtering. It is external to the canonical eOS80CH label source; molecule-level disjointness from the internal panel was not established for P3 and is not claimed. Feature extraction followed the identical pipeline used for the internal benchmark (ECFP4: Morgan radius 2, 2048 bits; TFP: 78 persistent homology features; TNE: 192-element Tucker core, bond dimension 8; external descriptor hybrid: TFP+TNE concatenation). The primary external analysis used intention-to-evaluate handling: TNE embedding failures ($n = 351$, 1.6 %) were retained as explicit zero-vector failure penalties rather than silently dropped. The complete-case sensitivity analysis ($n = 22\,096$) excluded those rows and gave ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798, versus ITT values of 0.960, 0.864, 0.645, and 0.800. The qualitative ranking was unchanged. Corrected-resampled tests are exploratory because the five folds have overlapping training sets; the naive fold-level p-values are shown only for comparison.

Table S14: *Descriptor external validation (5-fold Random Forest, ITT panel $n = 22\,447$; complete-case sensitivity $n = 22\,096$). The panel is external to the canonical eOS80CH label source, but molecule-level disjointness from the internal panel was not established for P3. Internal canonical values (5-fold CV, $n = 19\,836$) from Table 1 are shown for comparison. The ITT analysis retained embedding failures as explicit zero-vector penalties; the complete-case sensitivity excluded the same rows. AUC rankings were unchanged.*

Method	Internal AUC ($n = 19\,836$)	External AUC ($n = 22\,447$)	Δ
ECFP4	0.948	0.960	0.012
TFP	0.876	0.865	−0.011
TNE	0.722	0.645	−0.077
TFP+TNE	0.888	0.800	−0.088

19.2 Quantum kernel external validation

The quantum kernel benchmark was repeated on the same external ChEMBL panel (ITT $n = 22\,447$, 5-fold SVM with precomputed kernels) following the canonical protocol from section 18.1: UMAP reduction of 2048-bit Morgan fingerprints to 6 dimensions (Jaccard metric, `random_state=42`, $n_{\mathrm{neighbors}} = 15$, $min\_dist = 0.1$), scaling to $[-1, 1]$, and the 6-qubit IQPEmbedding state-vector kernel. The RBF gamma was tuned by inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$.

Table S15: *Kernel external validation (5-fold SVM, ChEMBL panel $n = 22\,447$). The panel is external to the canonical eOS80CH label source, but molecule-level disjointness from the internal panel was not established for P3. Internal values ($n = 19\,849$) from Table S13. QKS was lower than RBF externally (AUC 0.817 versus 0.847); the corrected-resampled comparison was exploratory ($p = 0.021$, BH $q = 0.021$), whereas the naive fold-level test gave $p = 0.005$.*

Kernel	Internal AUC ($n = 19\,849$)	External AUC ($n = 22\,447$)	p v
Quantum (IQPEmbedding, 6 qubits)	0.823	0.817	$p_{\mathrm{corr}} = 0.0$
RBF (gamma-tuned, SVM)	0.829	0.847	
Linear	0.537	0.544	$p <$

19.3 H_1–RRS association is size-mediated

The Spearman correlation between H_1 count and resistance-resilience scores (RRS) (Temgoua et al., 2026) was examined in two cohorts: (i) an expanded cohort of $n = 494$ compounds with complete H_1 and continuous RRS profiles (used for the cross-paper analysis in Figure S1), and (ii) a subset of $N = 77$ compounds with validated per-target RRS classification (used for the H_1–RRS association in section 19.3). In the expanded cohort ($n = 494$), H_1 count showed a weak unadjusted association with continuous RRS (Spearman $\rho = 0.239\,9$, $p = 6.76 \times 10^{-8}$), but the partial association was null after controlling for molecular weight, ring count, fraction sp^3, and H_0 count ($\rho_{\text{partial}} = 0.032\,9$, $p = 0.466\,8$). In the validated subset ($N = 77$), the raw correlation was $\rho = 0.312$ ($p = 0.006$) but dropped to $\rho_{\text{partial}} = 0.033$ ($p = 0.47$) after the same adjustment. Both analyses converge on the same conclusion: the H_1–RRS association is mediated by molecular complexity rather than topological structure per se. This result does not support a direct H_1–RRS association but is consistent with the broader finding that topological complexity correlates with binding promiscuity (section 3.6).

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